HARIKUMAR VIJAYKUMAR

M.Sc. Artificial Intelligence for Industrial Applications | Machine Learning & Computer Vision

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Profile

Master's student in AI for Industrial Applications at OTH Amberg-Weiden with hands-on experience in deep learning, computer vision, LLM-based retrieval (RAG) systems, and robotics. Builds end-to-end ML pipelines from data preprocessing through rigorous evaluation and deployment. Available as a Working Student (Werkstudent) 15-20 hr/week while enrolled full-time.

Technical Skills

- **Languages:** Python, C, C++, SQL
- **ML / DL Frameworks:** PyTorch, scikit-learn, XGBoost, sentence-transformers, NumPy, Pandas, feature engineering
- **Computer Vision:** OpenCV, image classification, anomaly detection (PatchCore), CNNs & Vision Transformers, data augmentation
- **LLM / GenAI:** RAG pipelines, embeddings, FAISS, Ollama (Llama 3.2), dense / hybrid retrieval, retrieval evaluation (Hit@k, MRR), LLM vision (Claude), attention mechanisms
- **ML Concepts:** Transfer learning, time series analysis, unsupervised learning, bootstrap confidence intervals
- **Tools & Practices:** Git/GitHub, FastAPI, Streamlit, Jupyter, LaTeX, VS Code

Projects

SentinelEdge — Two-Stage AI Inspector for Industrial Defect Detection *2026*

Personal Project — github.com/harigittihub/SentinelEdge-a-two-speed-AI-inspector

- Built a two-stage industrial inspection pipeline on MVTec AD: PyTorch PatchCore-style anomaly detection, then Jev (fast typed-triage layer) for triage decisions, then Claude vision for ambiguous cases; evaluated on a pre-registered held-out split with bootstrap confidence intervals.
- Cut false alarms from 11 to 6 and human-review load by 34% (86 to 57 images) with no additional missed defects observed; benchmarked Jev at 311 ms median latency (2.7x faster and about 20x cheaper than Claude Haiku 4.5).

PDF Question-Answering RAG System *2026*

Personal Project — github.com/harigittihub/RAG-System

- Built a PDF question-answering RAG system in Python (scikit-learn TF-IDF, sentence-transformers + FAISS, Ollama/Llama 3.2, Streamlit).
- Evaluated TF-IDF, dense and hybrid retrieval over chunk sizes on 30 hand-labeled questions (best: TF-IDF, 100-word chunks, Hit@5 0.97, MRR 0.78); added a citation check that flags pages cited outside the retrieved context.

Deep Vision Model Comparison — Comparative Study (5 architectures) *2025*

OTH Amberg-Weiden — Course Project

- Benchmarked ResNet18, MobileNetV3-Small, EfficientNet-B0, MobileViT-S, and DeiT-Tiny on an 8-class PlantVillage subset; MobileNetV3-Small gave the best accuracy-to-efficiency trade-off and MobileViT-S the top accuracy.

NASA Turbofan Predictive Maintenance — RUL Prediction *2025*

Personal Project — github.com/harigittihub/NASA-Turbofan-Predictive-Maintenance

- Built an end-to-end Remaining Useful Life (RUL) pipeline on NASA CMAPSS sensor data: preprocessing, label engineering, EDA, sliding-window sequences.
- Benchmarked an XGBoost regressor against an LSTM (PyTorch); deployed a Streamlit demo.

Unsupervised Object Sorting with a Niryo Robotic Arm *2025*

OTH Amberg-Weiden — Academic Project

- Combined a Niryo robotic arm, onboard camera vision, and K-Means clustering to autonomously sort objects by color within shape categories, integrating robotics, computer vision, and unsupervised ML.

Education

M.Sc. Artificial Intelligence for Industrial Applications *Oct 2024 – Ongoing*

OTH Amberg-Weiden | Coursework: Machine Learning, Deep Learning, Computer Vision, Industrial AI Systems, Time Series Analysis, Robotics

B.Tech. Computer Engineering *Aug 2019 – Jul 2023*

APJ Abdul Kalam Technological University, Kerala, India

Certifications

- Claude Code 101 & Claude Code in Action — Anthropic | Artificial Intelligence Foundations: Thinking Machines
- Software Engineering Lite Virtual Experience Program — Forage

Languages

English: Full Professional Proficiency (C1) | **German:** A2, actively progressing toward B1